

Laboratory 3: Process Scheduling

Simulation and Comparative Analysis of CPU Scheduling Algorithms

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Abstract

This report compares FIFO, Round Robin (RR), and Multi-Level Feedback Queue (MLFQ) scheduling using the recorded results for five workloads on one and two processors. The analysis focuses on turnaround time, the RR quantum, and periodic MLFQ priority boosts. For the two workloads used in the parameter studies, FIFO gives the lowest average turnaround on one processor. With two processors, RR with a quantum of 2 performs best on Workload 4, while FIFO remains best on Workload 5. Adding a second processor produces a larger improvement than parameter tuning in these two workloads, but does not uniformly halve turnaround time. Periodic boosting improves single-processor MLFQ averages by approximately 3%, whereas its two-processor benefit is workload-dependent.

1 Introduction

The objective of this experiment is to examine how CPU scheduling policies affect process completion under different workloads and processor counts. Part I compares FIFO, RR, and MLFQ on a single processor. Part II repeats the comparison with two processors sharing the ready queues.

The central questions are whether a smaller RR quantum improves average turnaround time, whether periodic MLFQ boosts help, and whether the ranking of policies changes when a second processor is available. The tables retain the recorded numerical results, while the five supplied figures visualize the parameter studies and processor-count comparisons.

2 Experimental Setup and Metrics

2.1 Workloads and scheduling policies

The experiments use five workload files, identified as Workloads 1–5. Each process has an arrival time and alternating CPU and I/O bursts. Workloads 1 and 2 contain pure-CPU jobs, while Workload 3 contains short jobs with I/O. Workload 4 contains three processes with several CPU/I/O cycles. Workload 5 contains six processes, four of which arrive together with identical burst patterns. FIFO is evaluated on all five workloads; the RR and MLFQ parameter studies focus on Workloads 4 and 5.

The scheduling configurations are:

- **FIFO:** a process keeps its CPU until its current CPU burst finishes; waiting processes are served in queue order.
- **RR:** a shared ready queue is time-sliced using quanta $q \in \{1, 2, 4, 8\}$ time units.
- **MLFQ:** three priority queues, each with a quantum of 2. New processes enter the highest-priority queue. A process that exhausts its quantum without completing its burst moves down one level, remaining at the lowest level once it reaches it. I/O completion returns a process to its existing level. The tested boost settings are no boost and a reset of all active processes to the highest level every 20 time units.

The simulation advances in discrete time units. With two processors, both draw from shared ready queues and execute concurrently in each tick; a single process cannot run on both at once. These details define the experimental model without requiring an implementation listing.

2.2 Interpreting the measurements

Turnaround time is the elapsed time from a process’s arrival to its final completion, including CPU execution, I/O, and ready-queue waiting. **Average turnaround time** is the main comparison metric, with lower values indicating earlier completion on average. **Maximum turnaround time**, reported for FIFO, captures the longest turnaround in each workload. All times are in simulated time units.

The original output’s “run time” is the sum of CPU-burst durations across all processes, excluding I/O. It is labeled **Total CPU work** here to distinguish it from elapsed workload completion time or the simulator’s execution time on the host computer. This quantity is unchanged when the same work is distributed over two processors.

Percentage reductions are computed from the displayed, rounded values as

$$\text{Reduction (\%)} = 100 \frac{\text{baseline average} - \text{comparison average}}{\text{baseline average}}.$$

The reported data do not quantify response time, context-switch cost, or fairness. Conclusions about these properties are therefore kept separate from measured turnaround-time improvements.

3 Part I: Single-Processor Results

3.1 FIFO baseline

Table 1 establishes the single-processor baseline.

Table 1: FIFO results, single processor. All values are in time units.

Workload	Avg. turnaround	Max. turnaround	Total CPU work
1	363.33	590	600
2	525.00	600	600
3	14.50	16	12
4	123.00	163	150
5	189.00	225	240

Workloads 1 and 2 have the same total CPU work, but their average turnaround times differ substantially: 363.33 versus 525.00. Thus, total CPU demand alone does not determine turnaround time; its distribution among processes and their arrival pattern also matter. Workload 3 has the smallest average and maximum turnaround times, consistent with its much smaller CPU demand.

For Workloads 4 and 5, the maximum turnaround exceeds the average by 40.00 and 36.00 time units, respectively. The recorded FIFO schedule for Workload 4 illustrates delayed service of short bursts: the first bursts of processes P1, P2, and P3 run at times 1–30, 31–40, and 41–43, respectively. P3 therefore waits behind longer bursts despite needing only three CPU units for its first burst. This demonstrates queueing delay, not evidence of indefinite starvation. Despite this limitation, FIFO achieves the lowest single-processor average among the tested policies and settings for both Workload 4 and Workload 5.

3.2 Round Robin: effect of time quantum

Table 2 and Figure 1 show the RR quantum sweep. Bold values identify the lowest average in each table row.

Table 2: RR average turnaround time, single processor.

Quantum	1	2	4	8
Workload 4	130.00	131.33	131.67	128.33
Workload 5	203.83	215.67	207.17	205.67

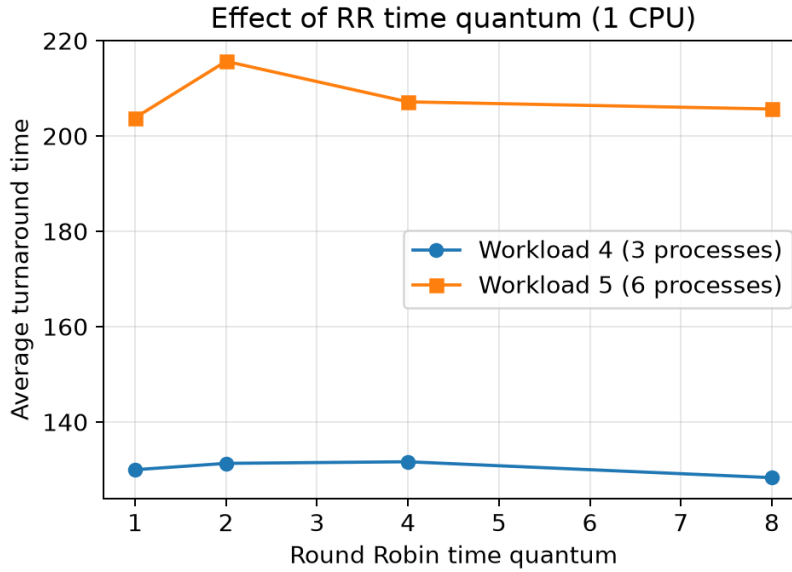


Figure 1: Single-processor RR quantum sweep for Workloads 4 and 5. The best tested quantum differs between the workloads.

Results. Workload 4 performs best at $q = 8$, with an average turnaround of 128.33, not at the smallest quantum. Its averages span only 3.34 time units across the sweep. Workload 5 performs best at $q = 1$ (203.83) and worst at $q = 2$ (215.67), a spread of 11.84 time units. Neither workload shows a monotonic improvement as the quantum increases. The results therefore do not support a universally best quantum.

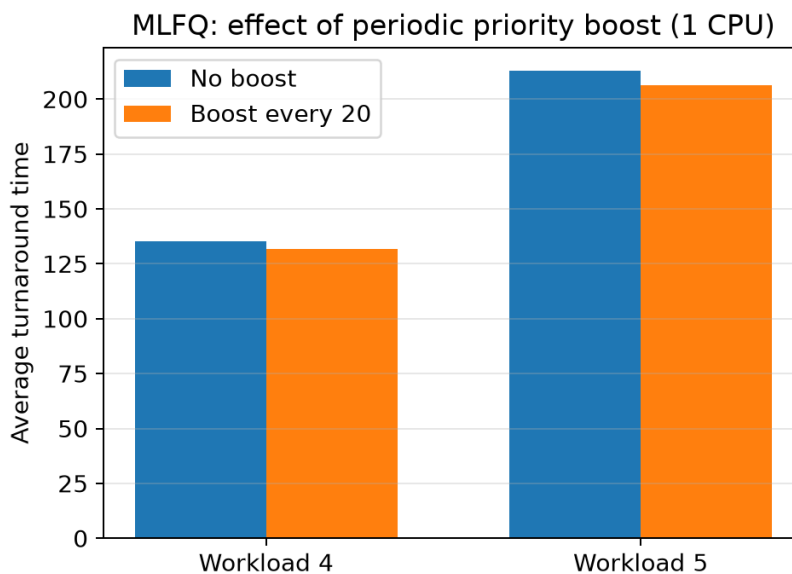
Even the best RR average exceeds FIFO by 5.33 time units on Workload 4 and 14.83 on Workload 5. Time slicing does not improve mean completion time for these single-processor workloads. It may offer benefits for other objectives, but response-time and fairness measurements would be needed to evaluate them. The observed differences should not be attributed to context-switch overhead, since no such cost is quantified in the results.

3.3 MLFQ: effect of priority boosting

The boost comparison is shown in Table 3 and Figure 2.

Table 3: MLFQ average turnaround time and reduction from boosting, single processor.

Workload	No boost	Boost every 20	Reduction
4	135.33	131.67	2.70%
5	212.83	206.17	3.13%

**Figure 2:** Single-processor MLFQ: boosting every 20 time units reduces average turnaround time for both workloads.

Results. Boosting reduces the average by 3.66 time units on Workload 4 and 6.66 on Workload 5. These are consistent but modest gains. Resetting priorities gives demoted processes renewed access to the highest queue, changing the order in which their remaining bursts are served. However, the averages alone do not establish which processes benefit or whether maximum turnaround time improves.

With boosting enabled, MLFQ still has a higher average than FIFO and the best tested RR setting on both workloads. Boosting improves MLFQ relative to its own unboosted baseline, but does not make it the best single-processor policy for average turnaround in this experiment.

4 Part II: Two-Processor Results

4.1 FIFO and the benefit of additional capacity

Table 4 compares two-processor FIFO with the single-processor baseline.

Table 4: FIFO results, two processors. Reduction is relative to the single-processor average in Table 1.

Workload	Avg. turnaround	Max. turnaround	Total CPU work	Reduction
1	230.00	390	600	36.70%
2	275.00	300	600	47.62%
3	13.50	14	12	6.90%
4	81.33	93	150	33.88%
5	104.33	119	240	44.80%

Every workload benefits, but the reduction ranges from just 6.90% on Workload 3 to 47.62% on Workload 2. Workload 3 shows that a second CPU does not necessarily halve turnaround time: short CPU demand and periods of I/O limit the amount of ready work that can execute concurrently. Workloads 4 and 5 improve by 41.67 and 84.67 time units, respectively. Their maximum turnaround times also fall, from 163 to 93 and from 225 to 119. Total CPU work remains unchanged for the same workloads.

4.2 Round Robin: reduced quantum sensitivity

Table 5: RR average turnaround time, two processors.

Quantum	1	2	4	8
Workload 4	81.67	79.67	81.67	81.67
Workload 5	107.50	107.33	106.83	105.67

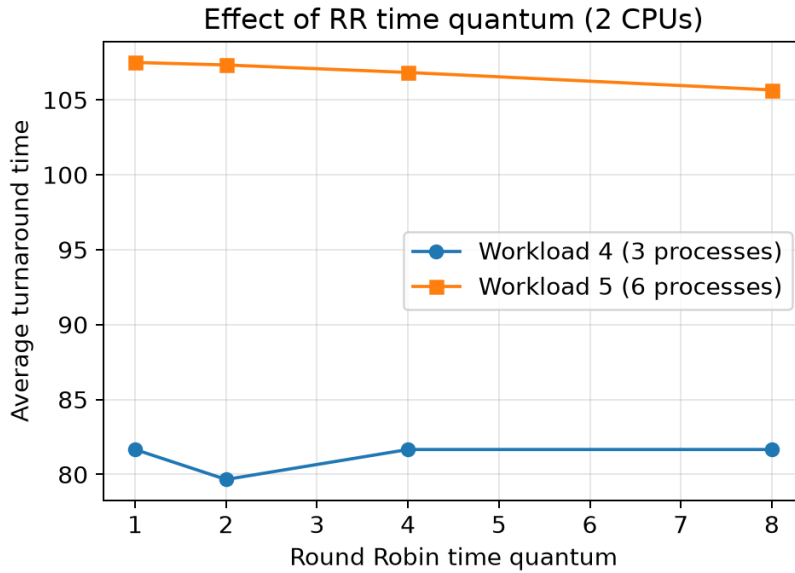


Figure 3: Two-processor RR quantum sweep. Workload 4 favors $q = 2$, whereas Workload 5 favors $q = 8$ among the tested settings.

Results. The best quantum changes from 8 to 2 for Workload 4 and from 1 to 8 for Workload 5 when moving to two processors. A quantum selected for one processor should therefore not be assumed optimal for two. The spread across quanta falls from 3.34 to 2.00 time units for Workload 4 and from 11.84 to 1.83 for Workload 5. These numerical spreads, rather than visual slopes alone, establish the reduced sensitivity because the two figures use different vertical-axis scales.

On Workload 4, RR at $q = 2$ achieves 79.67, improving on two-processor FIFO's 81.33 by 1.66 time units. On Workload 5, the best RR average is 105.67 at $q = 8$, still 1.34 time units above FIFO's 104.33.

4.3 MLFQ: boosting is not uniformly beneficial

Table 6: MLFQ average turnaround time, two processors. A negative reduction indicates that boosting worsens the average.

Workload	No boost	Boost every 20	Reduction
4	80.67	80.33	0.42%
5	107.50	110.17	−2.48%

Results. Boosting improves Workload 4 by only 0.34 time units but worsens Workload 5 by 2.67. The single-processor improvement therefore does not generalize to both two-processor cases. A priority reset can change queue order in ways that help some bursts and delay others; the aggregate results do not identify the precise per-process cause. Importantly, MLFQ demotion is triggered by quantum consumption, not by time spent waiting in a ready queue. More CPU capacity does not itself prevent demotion. The evidence supports workload-dependent boost tuning, not a claim that boosting always improves completion time.

5 Cross-Processor Comparison

Figures 4 and 5 compare fixed settings across processor counts. The supplied Workload 4 chart uses RR with $q = 2$, whereas the Workload 5 chart uses $q = 8$. Both use an MLFQ boost interval of 20. These are the settings plotted in the supplied images, not the best single-processor setting for every algorithm.

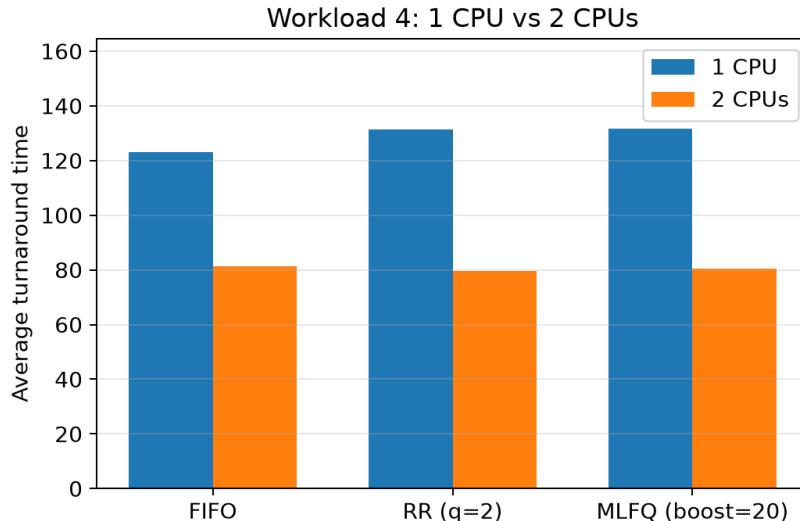


Figure 4: Workload 4: one versus two processors using FIFO, RR with $q = 2$, and MLFQ with a boost interval of 20.

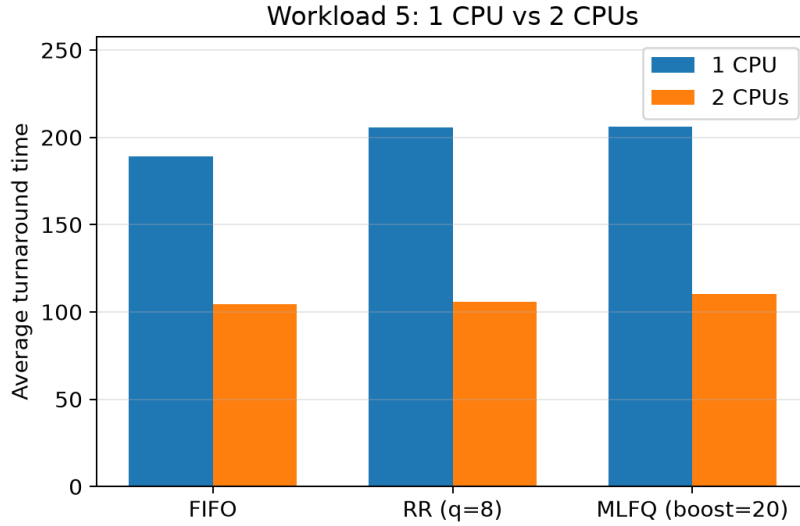


Figure 5: Workload 5: one versus two processors using FIFO, RR with $q = 8$, and MLFQ with a boost interval of 20.

Table 7: Numerical comparison for the configurations plotted in Figures 4 and 5.

Workload	Policy and setting	1 CPU	2 CPUs	Reduction
4	FIFO	123.00	81.33	33.88%
4	RR, $q = 2$	131.33	79.67	39.34%
4	MLFQ, boost every 20	131.67	80.33	38.99%
5	FIFO	189.00	104.33	44.80%
5	RR, $q = 8$	205.67	105.67	48.62%
5	MLFQ, boost every 20	206.17	110.17	46.56%

For these fixed configurations, the reductions range from 33.88% to 39.34% on Workload 4 and from 44.80% to 48.62% on Workload 5. RR has the largest percentage reduction in each chart, not FIFO. Workload 5 benefits more in percentage terms across all three plotted policies, consistent with its larger set of processes offering more opportunities for concurrent execution.

Processor count also changes the policy ranking on Workload 4: FIFO has the lowest single-processor average, but RR with $q = 2$ has the lowest two-processor average. For Workload 5, FIFO remains best even when RR and MLFQ are each evaluated at their best tested settings. Additional capacity has a much larger effect than the quantum or boost adjustments examined here, but no policy wins in every processor/workload combination.

6 Limitations and Scope

This analysis uses the recorded tables and supplied plots; it does not introduce new simulator measurements. Only five workloads are represented, and the parameter sweeps cover two of them, four RR quanta, and two MLFQ boost settings. “Best” therefore means best among those tested settings, not a global optimum or a general recommendation for operating systems.

Averages can conceal differences between processes. Only the FIFO tables report maximum turnaround, so the RR and MLFQ comparisons cannot establish improvements in worst-case completion time. Likewise, the results do not measure interactive response, fairness, or context-switch overhead. The unchanged total CPU work is not evidence of unchanged elapsed com-

pletion time, and a reduction in average turnaround should not be interpreted as a measured throughput speedup.

7 Summary of Findings

- **Single-processor ranking:** FIFO produces the lowest average turnaround for both detailed-study workloads: 123.00 for Workload 4 and 189.00 for Workload 5. Preemption does not improve this metric for these cases.
- **RR tuning:** the best tested quantum depends on both workload and processor count. It changes from 8 to 2 for Workload 4 and from 1 to 8 for Workload 5 when adding a second processor.
- **MLFQ boosting:** boosting every 20 time units improves single-processor averages by 2.70% and 3.13%, but gives only a 0.42% improvement on two-processor Workload 4 and a 2.48% deterioration on two-processor Workload 5.
- **Two-processor ranking:** RR with $q = 2$ is best on Workload 4 (79.67), while FIFO is best on Workload 5 (104.33), among the reported configurations.
- **Capacity versus policy:** adding a second processor reduces average turnaround in every reported paired comparison, but not uniformly by half. For Workloads 4 and 5, the capacity benefit outweighs the improvements from the parameter sweeps.

Overall, scheduling choices should be judged against the metric and workload of interest. In this experiment, a simple FIFO baseline is often competitive or best for mean turnaround, while RR and MLFQ require workload-specific tuning. Additional processor capacity changes both the size of the gains and, in some cases, the preferred policy.